nature communications

Article

https://doi.org/10.1038/s41467-025-62164-4

Self-adaptable mechanical ceramic fibrous aerogels from prestressed topology and multistable constraints

Received: 29 May 2024

Accepted: 11 July 2025

Published online: 25 July 2025

Check for updates

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Ceramic aerogels, widely used as thermal insulation materials, are renowned for their remarkable characteristics, including ultralight weight and ultralow thermal conductivity. However, their application is often limited by susceptibility to damage under repeated dynamic thermal shocks—a challenge that remains inadequately addressed. Herein, we present a multicomponent structural engineering approach that integrates ceramic nanofibers with traditional textile knitting topology to fabricate mechanically adaptable ceramic fibrous aerogels. Benefiting from the porous nanofibrous network and the synchronized motion of the prestressed knitted topological framework, which can be readily activated to accommodate deformation while efficiently dissipating energy, the resulting aerogels exhibit exceptional mechanical properties. Specifically, our aerogels demonstrate a high tensile strength of 356.6 kPa, a compressive strength of 109.1 kPa, and remarkable mechanical adaptability in response to external stimuli. Moreover, these aerogels achieve a high fracture energy of 117.26 kJ m^{-3} and display exceptional recovery from deformation after 1000 cycles of compression or 500 cycles of tension. This study elucidates the structural-property interdependence in aerogel materials through multiscale analysis and advances the rational design of the next-generation impact-absorbing systems and metamaterials.

Ceramic aerogels, characterized by ultralow density, ultrahigh porosity, and unprecedented thermal resistance, represent a promising class of "3U" thermal insulation materials suitable for a wide range of potential applications, such as aerospace, national defense, and the energy sector[1–8]. Nevertheless, the ongoing exploration of deep sea and aerospace environments imposes increasing demands on the mechanical properties of ceramic aerogels[9]. Conventional ceramic aerogels, typically characterized by an intrinsic "pearl necklace" structure, tend to exhibit brittleness[9–11]. Therefore, significant modifications have been implemented to improve the insufficient mechanical performance of the ceramic aerogels by incorporating continuous building blocks such as nanofibers and nanowires to develop unique three-dimensional (3D) cellular structures (SiO_2 nanofibrous aerogels[12–15], mullite nanofibrous aerogels[16], and SiC@SiO_2 nanowire aerogels[17,18]). As expected, the high aspect ratio of the building blocks and the unique microstructure contribute to the exceptional compressive strength and flexibility of these aerogels; however, their tensile performance remains as a critical limitation[10–18]. Further advancement involves the direct electrospinning of ceramic nanofibrous aerogels with 3D interwoven curled nanofiber structures, which exhibit a modest enhancement in tensile strength (up to 12.7 kPa)[19]. More recently, we have explored the regulation of stress diffusion and crack propagation pathways through Bouligand-chiral templated assembly combined with microfiber-based mechanical reinforcement, yielding ultra-strong ceramic fibrous aerogels with a tensile strength of 170.38 MPa[9]. Indeed, the effects of mechanical

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adaptation are often overlooked, yet they may hold even greater significance—enabling aerogel materials to autonomously modulate their mechanical behavior in response to external stimuli via mechanically activated processes. This capability allows for preferential load redistribution and energy dissipation, thereby reducing the risk of failure in ceramic aerogels. However, the absence of multiscale structural control—ranging from customizable macroscopic architectures to well-ordered microscopic configurations—remains a significant challenge in developing ceramic fibrous aerogels that exhibit both superior mechanical robustness and adaptability[20].

Knitted fabrics, distinguished by their dynamic mechanical properties, have attracted considerable attention in recent years[21,22]. Knitting is characterized by a distinct network of repeated stitches made from a single yarn, creating a structure that is constrained to self-crossing topological loops (i.e., topological units), which remain conserved under external stimuli[22]. Critically, these loops, due to their inherent curvature, can undergo significant deformation, allowing for the dynamic adjustment of yarn positioning between adjacent stitches in response to external forces. This property enables knitted structures to exhibit mechanical adaptability, effectively mitigating stress concentrations[21–25]. Building on these features, we synergize knitted topological networks (KTNs) design with ceramic nanofiber aerogels assembly to explore the self-adaptable mechanical properties of aerogels. Some research has investigated the basic principles of geometric and topological dynamics in KTNs. For instance, the modifications of Euler–Bernoulli beam theory models were presented to analyze the configuration characteristics of stitches in KTNs[26–29]. Additionally, an approach has been developed that conceptualizes KTNs as a framework in which stitches serve as fundamental units, allowing for control of the overall deformation behavior through the setting of boundary conditions or the coupling of kinematics among neighboring units[30,31]. These pioneering efforts established the foundation and theoretical framework for engineering the self-adaptable mechanical ceramic fibers-based aerogels.

Herein, we present a multicomponent structural engineering strategy that integrates an ancient textile technique—knitting—with ceramic nanofibers to design and fabricate a type of ceramic fibrous aerogel-based mechanical and adaptable metamaterials, termed mechanically adaptable fibrous aerogels (MAFAs). Our approach employs a multiscale strategy that combines micro- and nanoscale control with precisely defined structural degrees of freedom, thereby enhancing the efficiency of topological unit activation. The resulting MAFAs demonstrate intriguing dynamic and robust mechanical properties. Moreover, this study incorporates mechanical experimental characterization with multiscale mechanical simulations to systematically investigate the enhancement mechanisms underlying MAFAs. The theoretical and simulation modeling framework developed herein holds promise for extension to other fibrous aerogel systems and 3D materials.

Results

Formation of prestressed topology and self-adaptable mechanical MAFAs

As a proof-of-concept study, we used a ceramic plied yarn and a computerized flat knitting machine (LXA252SCV-7G) to fabricate KTNs with a topologically basic stitch pattern, which served as a built-in topological framework in the MAFAs (Supplementary Fig. 1)[21–25]. In particular, the ceramic yarns are composed of continuous filament ceramic oxide microfibers, which have significant merits, such as high-temperature resistance (~1300 °C), flexibility and robustness, and textiles that can be manufactured directly away from any other fiber or wiring inlays.

The fabrication of MAFAs was achieved through an integrated process combining freezing-drying with prestressed structural assembly, as illustrated in Fig. 1a and Supplementary Fig. 2. The synthesis began with the preparation of a silane sol via controlled hydrolysis-polycondensation employing tetraethyl orthosilicate (TEOS), deionized water, ethyl alcohol, and phosphoric acid as components (Supplementary Fig. 3). The electrospun SiO_2 ceramic nanofibers (Supplementary Fig. 4, preparation process see Methods)[10] were then uniformly dispersed into the silane sol matrix through ultrasonic processing to create a homogeneous nanofibers/sol dispersion. The KTN framework was tensioned to a certain prestrain and immersed in the nanofiber/sol dispersion for 1 h (Fig. 1a i and Fig. 1b), allowing for complete infiltration of the porous structure. Rapid cryogenic fixation at liquid nitrogen (−196 °C) preserved the aligned microstructure while inducing directional ice crystal growth (Fig. 1a ii and Fig. 1c), followed by a 48-h freeze-drying process to remove the ice templates. Final consolidation was achieved through calcination process, which simultaneously eliminated organic components and established covalent Si-O-Si bridges between adjacent fibers (Fig. 1a iii and Fig. 1d).

There are two structural features with different scales that are intertwined in MAFAs. As depicted in the SEM image in Fig. 1e, the unique multiscale structure of MAFAs consists of the KTNs framework and the homogenous interconnected nanofibrous porous network. Mechanistically, the formation of the nanofibrous porous structure is a consequence of phase separation resulting from the initiation of water crystallization[32]. During freezing, the phase separation process resulted in the repelling and accumulation of solid components of the nanofibers in different directions around the developing ice crystals. This process allowed the nanofibers to form a continuous and porous network between the crystals, which then enveloped the KTNs framework of the ceramic microfibers. After that, the silane sol congregated onto the surface of the fibers, forming an elastic and homogeneous Si-O-Si cross-linking network and intertwining along with the condensation of the TEOS. Following freeze-drying, the ice crystal template was removed, causing the ice crystals to transition to a gaseous state[12]. Moreover, the nanofibers framework remained intact, resulting in the formation of a porous network structure. Calcining at a high temperature formed robust bonding (Fig. 1h, i).

High-resolution scanning electron microscopy (SEM) characterization reveals the distinct micro/nano-scale composite structural evolution mechanism of MAFAs. Detailed SEM analysis identifies a dual-network architecture consisting of: (i) a primary structural framework composed of ceramic microfibers arranged through KTN configurations, and (ii) a secondary reinforcement network formed by uniform SiO_2 nanofibers via ice-templated self-assembly. Crucially, the residual stress field introduced by the prestressed KTN framework induces a contracted configuration mode in the 3D porous nanofiber networks (Fig. 1f, g), thereby conferring a high capacity for elastic potential energy storage. Notably, the engineered prestress fundamentally enhances mechanical resilience through localized stress redistribution[11]. This coordinated mechanical response arises from multiscale stress synchronization between the ceramic microfiber-based KTNs framework and the nanofibrous porous network, thereby improving strain energy dissipation efficiency, an essential mechanism that underpins the remarkable mechanical adaptability of the MAFAs, such as ultrahigh strain recovery and rapid shape responsiveness[33,34].

Mechanical mechanism based on multistable topological constraints

Figure 2a and Supplementary Fig. 5 illustrate the absolute reference configuration and the corresponding deformed state of the KTNs. In this structure, the stitches are arranged into rows and columns, referred to as the course and wale directions, respectively. Each topological unit of the KTNs consists of a single ceramic fiber yarn that alternately interweaves with adjacent top and bottom rows as it traverses each stitch row. During mechanical loading, the KTNs were tensioned in the wale direction, with clamps securing the top and bottom rows to prevent the displacement of the stitches, thereby

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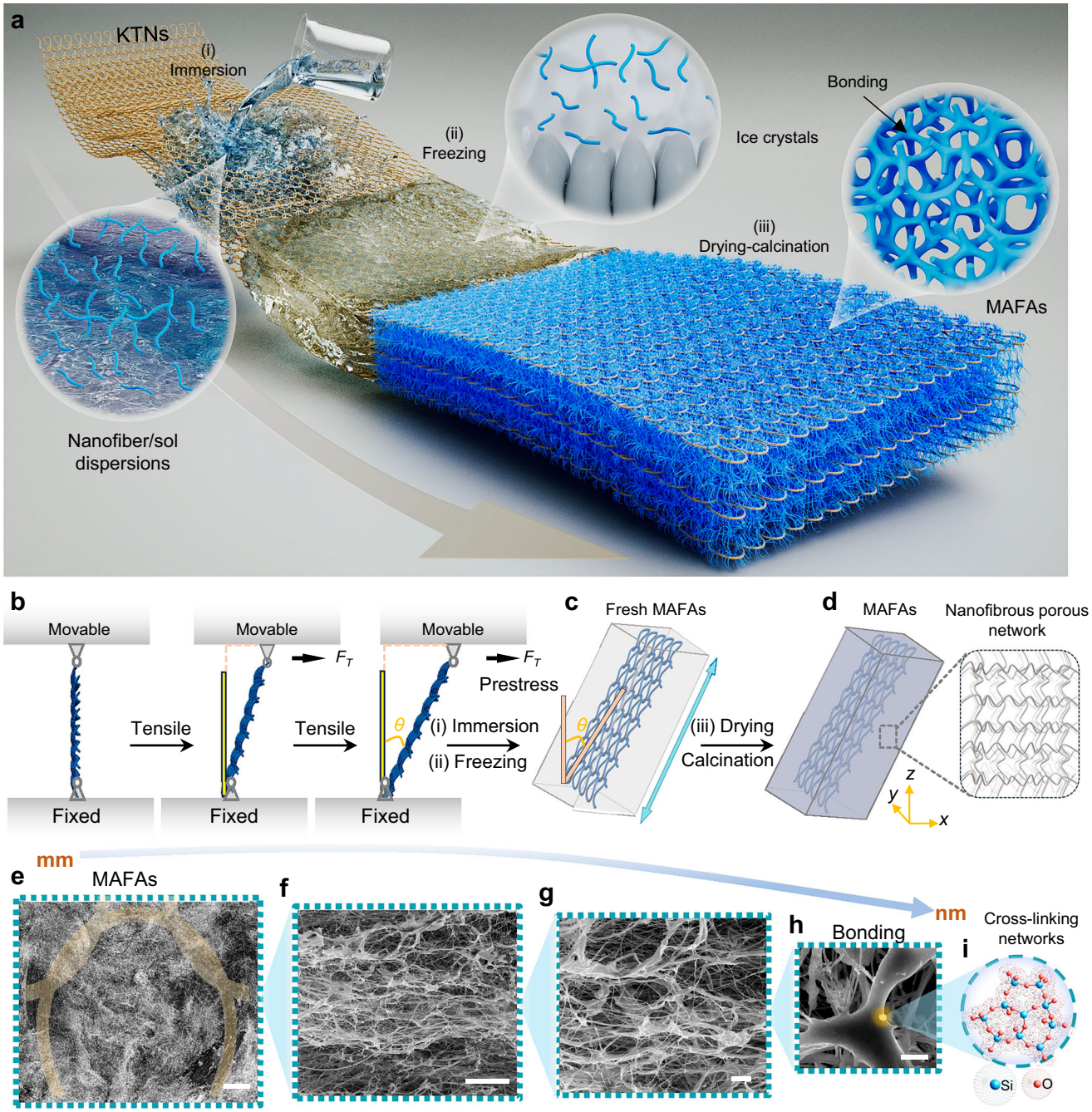


Fig. 1 | Structural design of the mechanically adaptable fibrous aerogels (MAFAs). a Schematic illustration of the fabrication process of MAFAs, highlighting the integration of prestressed knitted topological networks (KTNs) with ceramic nanofibrous aerogels through a freeze-drying-assisted assembly strategy. **b** A tensile force (F_T) was applied to the KTNs, inducing prestress within the framework. **c** The prestressed KTNs were rapidly frozen within the nanofiber/sol dispersions using liquid nitrogen. **d** The resulting MAFAs were prepared by calcining them in a muffle furnace at 800 °C. **e**–**h** Representative scanning electron microscopy (SEM) images showing the microscale structures of MAFAs at different significance levels. Yellow highlights of **e**: Schematic illustration indicating the positions of KTNs. Scale bars: **e** 500 μm; **f** 50 μm; **g** 10 μm; and **h** 5 μm. **i** The cross-linking networks among the adjacent fibers of the MAFAs.

ensuring $L_c = L_{c\text{-}initial}$, where $L_{c\text{-}initial}$ and L_c represent the initial and current lengths along the course direction, respectively[21–25]. Accordingly, the total strain of the KTNs during deformation is defined as:

$$\varepsilon_{total} = (L_w - L_{w-initial})/L_{w-initial} \quad (1)$$

where $L_{w\text{-}initial}$ and L_w denote the initial and current lengths along the wale direction, respectively.

The geometric configuration of the KTNs is described by connecting the geometric centers of adjacent stitches along both the wale and course directions. Furthermore, the yarn self-crossing topology of KTNs is represented as a four-degree planar graph, which is visually depicted by the light-gray lines in Fig. 2a. The graph is uniquely determined by the boundary vectors $\vec{\mathbf{m}}^*$ in the course direction and $\vec{\mathbf{n}}^*$ in the wale direction. In addition, their corresponding norms $m^* = \|\vec{\mathbf{m}}^*\|$ and $n^* = \|\vec{\mathbf{n}}^*\|$, define the local dimensions of the units in the absolute reference state, as shown in Fig. 2a i[22,23,28,30,35,36]. In the deformed state (Fig. 2a ii), the corresponding vectors are denoted as $\vec{\mathbf{m}}$ and $\vec{\mathbf{n}}$, with norms $m = \|\vec{\mathbf{m}}\|$ and $n = \|\vec{\mathbf{n}}\|$, respectively. Notably, both the directions and norms of these vectors in the deformed state

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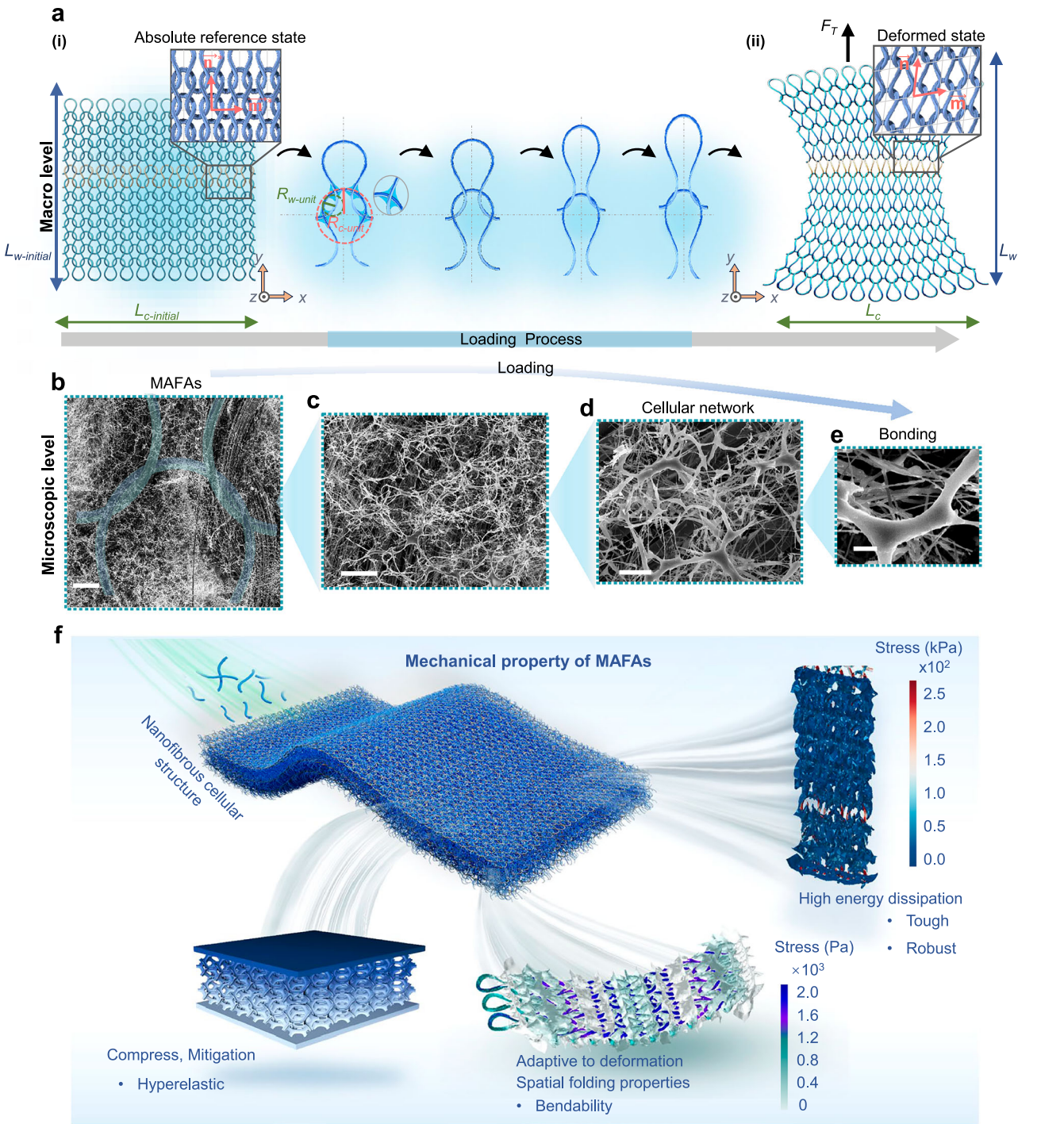


Fig. 2 | Mechanical mechanism based on multistable topological constraints. **a** Schematic diagram of **i** the reference configuration and **ii** the deformed configuration of knitted topological networks (KTNs), displaying the topology of the stitches and their arrangement in the representative catenary structure. (insets of **i**) Absolute reference state of the KTNs: The two vector fields $\vec{\mathbf{m}}$ and $\vec{\mathbf{n}}$ orthogonal and equivalent for each unit. (insets of **ii**) Deformed state of the KTNs: The directions and norms of $\vec{\mathbf{m}}$ and $\vec{\mathbf{n}}$ change over the KTNs. The global dimensions of the KTNs are indicated, along with the applied tensile loading direction. $L_{c\text{-}initial}$ and L_c denote the initial and current lengths along the course direction (horizontal axis), while $L_{w\text{-}initial}$ and L_w represent the initial and current lengths along the wale direction (vertical axis). $R_{w\text{-}unit}$ and $R_{c\text{-}unit}$ indicate the two characteristic radius of curvature of the yarn loops in the KTNs, which govern their deformation behavior under stress. F_T, the total force exerted on the KTNs. Macro level: the bulk-scale (millimeter to centimeter) structural and functional properties of the MAFAs. **b**–**e** Representative SEM images showing the microscopic structures of MAFAs at various magnifications following external loading. Blue highlights of **b**: Diagram indicating the spatial distribution of KTNs. Scale bars: **b** 500 μm; **c** 100 μm; **d** 20 μm; **e** 5 μm. **f** The structure-property relationships of the resulting MAFAs.

vary spatially throughout the KTNs, reflecting the geometric adaptability of the KTNs material. Moreover, the yarns are not rigidly fixed to the topological units, thereby allowing them to slide freely between stitches during deformation. Despite this mobility, the crossing points of the yarns remain invariant and cannot be interchanged. This is because the KTNs are governed by intrinsic topological constraints, which, in turn, impose the following kinematic condition:[22–24,31,36]

$$\vec{\mathbf{n}}(i,j)+\vec{\mathbf{m}}(i,j+1)=\vec{\mathbf{n}}(i+1,j)+\vec{\mathbf{m}}(i,j) \tag{2}$$

In Eq. 2, the indices (i, j) represent the stitch position relative to the course and wale directions, respectively; the vector field $\vec{\mathbf{v}}$ is a unified reference point with respect to the (x, y) plane. Additionally, KTNs maintain structural length conservation, and the yarn length in each unit is proportional to $l_{unit} = m + \delta n$, where δ quantifies differential yarn length contributions to stitch extensions along the course and wale directions[22,23,25,27].

In the (x, y) plane, two distinct curvature radii—$R_{c\text{-}unit}$ and $R_{w\text{-}unit}$—are introduced to characterize the local geometric curvature along the course and wale directions, respectively (Supplementary Fig. 6). Assuming that $R_{c\text{-}unit} \sim m$ and $R_{w\text{-}unit} \sim n$, the bending energy associated with deformations in the wale and course directions can be approximated as $E_w \sim M/n$ and $E_c \sim M/m$[23,24,31,36], where M represents the bending modulus of the ceramic fiber yarns. In the two-dimensional framework, the total elastic energy of each stitch can thus be estimated as $E_{k\text{-}elasity} = E_c + E_w$. Finally, by accounting for scale separation between the diameter of the ceramic fibrous yarn and the stitch extensions, and drawing on the theoretical framework established by Poincloux et al.[23], the Lagrangian $\mathcal{L}\{\vec{\mathbf{m}}, \vec{\mathbf{n}}\}$ of the KTNs, as illustrated in Fig. 2a, can be formulated in a continuous representation as[22–24,26,29]:

$$\mathcal{L} = \tilde{Y} \iint_{x,y} dxdy\left(\frac{1}{m} + \frac{\vartheta}{n}\right) - \iint_{x,y} dxdy\,\vec{\mathbf{T}}(x, y) \cdot \left(\frac{1}{m^*}\frac{\partial \vec{\mathbf{m}}}{\partial y} - \frac{1}{n^*}\frac{\partial \vec{\mathbf{n}}}{\partial x}\right) + \alpha \iint_{x,y} dxdy(m + \delta n) - \iint_{x,y} dxdy\mathcal{T}(x)\vec{\mathbf{e}}_y \cdot \vec{\mathbf{n}} \quad (3)$$

where $\tilde{Y}$ is effective tensile modulus, ϑ indicates a dimensionless parameter that quantifies the curvature asymmetry of topological unit, $\vec{\mathbf{T}}(x, y)$ is the specific stress at a given point, α represents a universal scaling factor for the tension in the KTNs, $\mathcal{T}(x)$ is a traction distribution applied to KTNs. Additionally, in Eq. 3, the first term represents the total elastic energy of the KTNs, the second term enforces local topological constraints, the third term maintains yarn-length conservation, and the final term accounts for the work of traction due to the applied external forces[22,23].

The findings indicate that KTNs have properties similar to those of rubber-like materials, being highly elastic. Simultaneously, in the event of heterogeneous deformation in the KTNs, the solution to Eq. 3 offers us information regarding differential equations for which the solutions guarantee energy minimization and satisfaction of the constraints. Overall, the intrinsic flexibility of the ceramic yarns, combined with the spatially deformable characteristics of the topological units within the KTNs, enable fiber-scale mobility of the MAFAs under external loading. Meanwhile, the interlocked architecture within the KTNs serves as a mechanical fulcrum, maintaining the directional responsiveness and mechanical selectivity of the system.

Subsequently, the structure-property relationships of both the cellular network and KTNs within the composite MAFAs were systematically investigated. Under tensile loading, the MAFAs exhibited a pronounced and coordinated structural reorganization. This response was characterized by the simultaneous realignment of the prestressed KTNs and the initially contracted nanofibrous cellular network, both of which progressively oriented along the direction of the applied tensile force. This suggests a synergistic mechanism intrinsic to the MAFA architecture, wherein the dynamically reoriented deformation of the KTNs directly couples with—and likely facilitates—the gradual straightened of the nanofibrous network (Fig. 2b–e). Such coordinated behavior is critical for ensuring the efficient transduction of mechanical loads throughout the entire structure, thereby enabling the MAFAs to adapt to deformation while effectively enhancing energy absorption, impact mitigation, and mechanical property regulation. More critically, the nanofibrous cellular architecture formed by cross-linked nanofibers facilitates a multistage, indeterminate deformation process under compression—comprising elastic distortion of the nanofiber cellular network, progressive fiber bending, and eventual failure at bonding points. This hierarchical deformation sequence collectively endows the MAFAs with outstanding compressive elasticity and superior energy dissipation capacity[10,21,37,38].

As shown in Fig. 2f, the resultant MAFAs synergistically integrate the intrinsic ultra-strength of the constituent ceramic fibers, the dynamic mechanical adaptability conferred by the KTNs, and the stress-dissipating capacity of the porous nanofibrous cellular architecture. This unique multiscale integration imparts the MAFAs with a suite of exceptional mechanical properties, including high toughness, enhanced tensile strength, superelasticity, and remarkable flexural resilience. Moreover, the modular and reconfigurable nature of their design affords broad versatility for the development of advanced, architected mechanical metamaterials[21–24,39,40].

Tensile property and mechanics simulations

The resulting MAFAs showed superior yet distinct mechanical behaviors, including super strength, high ductility and toughness along the KTN alignment direction, coupled with superelasticity, bendability and foldability perpendicular to the KTN orientation. Initially, the tensile mechanical behaviors of the MAFAs were characterized in detail via an Instron 34 TM-5 Universal Testing Machine. As shown in Fig. 3a, b, the MAFAs exhibited strong mechanical properties, with a toughness of 117.26 kJ M^{-3} and yield strength (σ_y) of 356.68 kPa, which is two orders of magnitude greater than that of the pure nanofiber aerogels (p-NFAs) (σ_y of 2.42 kPa and toughness of ~0.05 kJ M^{-3}). Mechanistically, this behavior arises from the unique dynamic coupling between the 3D nanofibrous porous network and the KTN framework. Originally, stress concentrations are primarily localized within the nanofibrous cellular domains, initiating progressive microcrack propagation and structural failure. Subsequently, the KTN topological units are activated, transitioning into a cooperative load-bearing regime through interfacial stress transfer mechanisms. The transferred stress reaches magnitudes comparable to the failure threshold of the nanofibrous structures, thereby inducing further localized fracture within the network[2]. Notably, rather than experiencing conventional brittle fracture behavior, the KTNs in MAFAs demonstrated sequential failure characterized by a distinctive sawtooth-patterned stress reduction (Fig. 3ai–v). This phenomenon originated from the unique stress-transfer mechanism of topological interlocking within the KTNs. Upon rupture of the primary topological unit by external loading, the interlocking configuration became destabilized, triggering progressive domain unfolding. This structural rearrangement enabled the release of previously constrained "hidden length" through an "unchaining" process and generated a jagged mechanical response—a highly energy-dissipative process while effectively mitigates stress concentration[41,42]. Final failure of the MAFAs occurred only when the external force exceeded the yield strength (σ_y) of the KTNs.

In stark contrast, the p-NFAs fractured directly without undergoing any observable stretching, underscoring their intrinsically poor toughness (inset in Fig. 3a and Supplementary Fig. 7). To elucidate the underlying mechanical mechanisms, finite element (FE) analysis was conducted under uniaxial loading conditions, with each structural component of the specimen explicitly modeled[43,44]. Fig. 3c and Supplementary Fig. 8a present the von Mises stress isosurface contours and stress distribution of the mechanical MAFAs under failure load. Notably, the FE simulation results exhibit excellent agreement with the experimentally measured elongation at break.

Moreover, the durability of self-adaptable mechanical materials is a critical consideration for aerospace thermal insulation applications[45]. To assess the fatigue properties of MAFAs, a 500-cycle tensile test with a ε of 30%, applied parallel to the KTNs, was conducted (Fig. 3d and Supplementary Figs. 9, 10)[45]. Remarkably, even after 500 loading-unloading cycles, no significant degradation in ε or occurrence of rupture was observed. The maximum stress (σ_{max}) remained at approximately 85% of that in the initial cycle (Supplementary Fig. 11). Concurrently, the consistent energy loss coefficient (~ 0.64) and stable

https://doi.org/10.1038/s41467-025-62164-4

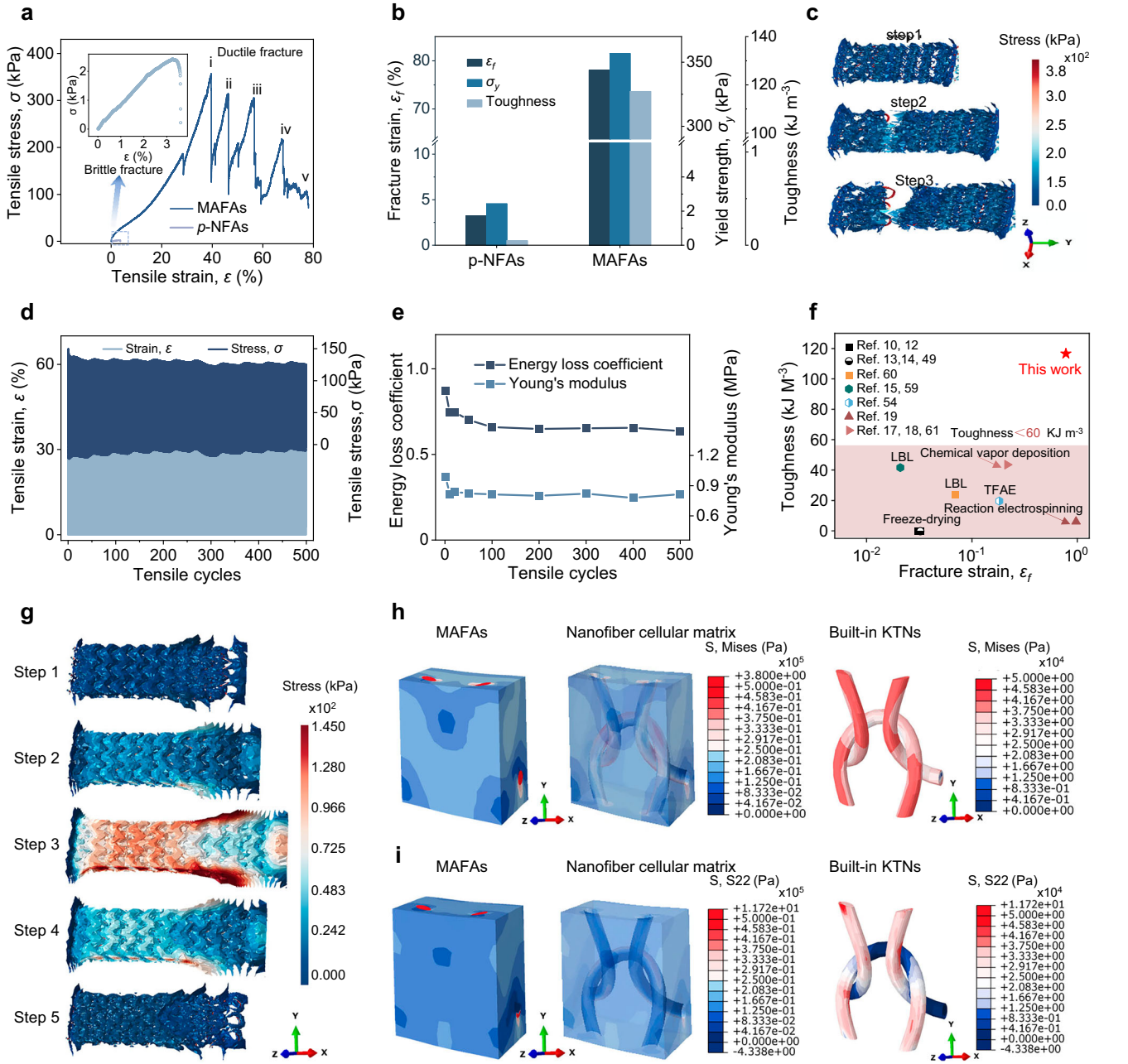


Fig. 3 | Tensile properties of MAFAs and mechanics simulations. a Tensile stress-strain (σ–ε) curve of MAFAs. (inset of **a**) Tensile σ–ε curve of the pure nanofiber aerogels (*p*-NFAs). Panel **i**–**v** of **a** indicate the distinctive sawtooth-patterned stress reduction by release the "hidden topological length" when MAFA under external force. **b** Comparison of the yield strength (σ_y), fracture strain (ε_f), and toughness of MAFAs and *p*-NFAs. **c** FE simulations depicting the tensile stress isosurface contours corresponding to the fracture mode of the MAFAs. **d** Tensile test at $\varepsilon = 30\%$ for 500 loading-unloading cycles. **e** Energy loss coefficient and Young's modulus versus number of stress cycles. **f** Comparison of the toughness and ε_f of MAFAs and other materials, including SiO_2 ceramic aerogels prepared by the freeze-drying method[10,12–14,48], SiO_2-ZrO_2 nanofibrous aerogels prepared by the layer-by-layer (LBL) method[15,57,58], zircon nanofibrous aerogels (ZAGs) prepared by the turbulent flow-assisted electrospinning (TFAE) method[53], SiC-SiO_2 aerogels prepared by the chemical vapor deposition method[17,18,59] and mullite aerogels prepared by the direct reaction electrospinning[19]. The shaded region represents the materials with toughness values below 60 kJ m^{-3}. **g** FE simulations illustrating the stress isosurface contours during tensile cycling of the MAFAs. **h, i** Simulation of the stress contours of the MAFAs, nanofiber cellular matrix, and built-in KTNs. S, Mises: von Mises stress of materials; S, S22: the normal stress of the x–z plane along the y axis. Source data are provided as a Source Data file.

Young's modulus throughout cyclic testing underscore the superior long-term mechanical stability of MAFAs (Fig. 3e)[19,45,46]. Importantly, the toughness of the developed MAFAs substantially exceeds that of previously reported ceramic aerogels (Fig. 3f).

To gain deeper mechanistic insights into the exceptional tensile cyclic behavior of MAFAs, FE simulation was performed to explicitly capture their mechanical response under cyclic loading conditions. Specifically, as shown in Fig. 3g and Supplementary Fig. 8b, the simulation results reveal a pronounced stress differential that unequivocally substantiates the dual-mechanism reinforcement strategy: the KTNs function as the primary load-bearing framework, establishing continuous stress-transfer pathways through topological unit constraints and interlocking, whereas the nanofibrous cellular structures undergo sequential buckling and collapse to dissipate energy[44]. This synergistic stress partitioning, as revealed by the FE simulations, aligns with the experimentally observed mechanical

enhancements—most notably, the substantial improvement in fracture toughness compared to *p*-NFAs.

On the other hand, mesoscale theoretical simulations were further conducted to elucidate the fundamental strengthening and toughening mechanisms underlying the performance of MAFAs[47]. As previously discussed, the observed enhancements in strength and fatigue resistance are primarily attributed to the integrated prestressed KTNs and the nanofibrous cellular architecture[47]. Therefore, leveraging the FE method for mesoscopic composite mechanics, a representative volume element (RVE) model was established, wherein periodic boundary conditions were applied to analyze the intrinsic mechanical properties of the MAFAs[37,47]. Due to the intrinsic heterogeneity in elastic modulus within the RVE model, the simulation results exhibited a characteristically nonuniform displacement field distribution (Supplementary Fig. 12). Figure 3h shows the von Mises stress distributions contours for the complete MAFAs, and the individual components: nanofiber cellular matrix and the built-in KTNs. The stress distribution within the nanofibrous matrix remained relatively uniform due to the low-modulus nature of the nanofibers and the isotropic cellular architecture, which collectively facilitated efficient stress transmission across the network. In contrast, the overall mechanical response of the MAFAs was predominantly governed by the prestressed KTNs, which carried most of the applied load while effectively constraining matrix deformation to below the nominal strain threshold. This stress partitioning strategy allows for the simultaneous optimization of strength and toughness. Specifically, the KTNs preserve structural integrity through their high-stiffness framework and topologically enabled energy storage capacity, whereas the nanofibrous matrix enhances energy dissipation via controlled cellular collapse and interfacial sliding.

Furthermore, the FE simulation results presented in Fig. 3i illustrate the distribution of the stress component (S22) within the MAFAs, the nanofibrous cellular matrix and the embedded KTNs. Importantly, the MAFAs demonstrated the capacity to withstand not only tensile but also compressive loads, which can be attributed to the mismatch in Poisson's ratio between the ceramic microfibers and the nanofibrous matrix. This mismatch facilitated a highly coordinated deformation response across the composite structure. More critically, the stress distribution within the KTNs appeared relatively nonuniform, a consequence of the mechanical coupling between the KTNs and the surrounding nanofibrous aerogel matrix. This interaction effectively constrained fracture propagation within the nanofiber matrix and played a crucial role in preserving the structural stability and morphological integrity of the MAFAs under mechanical loading.

Compression and bending performance

Following the characterization of tensile responses, multiscale analysis was conducted to elucidate the compressive strength and flexural performance of the MAFAs, revealing their unique stress mechanisms under complex loading conditions.

Cyclic compressive loading-unloading experiments were performed using dynamic mechanical analysis (DMA) under strain-controlled mode ($\varepsilon = 20–90\%$). As evidenced in Fig. 4a, the resulting σ–ε profiles demonstrated that the MAFAs achieved a maximum compressive stress (σ_{max}) of 109.1 kPa at 90% strain, revealing their remarkable deformability under large-strain conditions. Subsequently, detailed evaluations of the compressive characteristics of the MAFAs were conducted over 1000 loading-unloading cycles at a strain of 50%. As expected, the MAFAs exhibited no significant degradation in compressive performance, with only negligible plastic deformation observed even after extensive cyclic loading (Fig. 4b, c), revealing the exceptional fatigue resistance in the MAFAs[45]. More strikingly, despite rigorous testing, the MAFAs maintained a virtually identical energy loss coefficient and an almost unchanged Young's modulus from 10 to 1000 cycles (Fig. 4d). Additionally, the MAFAs demonstrated exceptional compression-recovery durability, maintaining remarkable structural integrity after 100,000 compressive cycles ($\varepsilon = 3\%$, $f = 5$ Hz). Only a slight decrease was observed in either the storage modulus or the loss modulus, while the damping ratio exhibited a steady-state value of approximately 2.5 (Fig. 4e)[13]. Furtherly, the dynamic compressive behavior of MAFAs was studied. As shown in Fig. 4f, both the storage modulus and the loss modulus remained constant over the angle frequency in the region from 0.01 to 100 Hz. The damping ratio also remained stable (~0.14) over the whole angular frequency range tested, suggesting the superior elastic response and fatigue resistance of the MAFAs[48]. In addition, optical photographs (Supplementary Fig. 13) captured the exceptional superelastic behavior of the MAFAs, as evidenced by the rapid rebound of a steel ball upon impact.

To further elucidate the bending behavior and the role of each component in the synergistic mechanical properties of MAFAs, a series of three-point flexural tests were performed utilizing an Instron 34 TM-5 Universal Testing Machine. As shown in Fig. 4g, the MAFAs exhibited superior ductility and flexural strength while maintaining their original geometric morphology even when subjected to a large diametral deflection of 38 mm in a three-point bending load applied with a 40 mm fixture span; however, no fracture was observed[49]. In sharp contrast, the nonprestressed aerogels (NP-MAFAs) and *p*-NFAs fractured at the mid-span position where the bending moment was the largest (insets in Fig. 4l and Supplementary Fig. 14). This was a consequence of the stress concentrating specifically at the crack tip as the load intensified, ultimately leading to rapid crack propagation and rupture[50]. The σ–ε curves in Fig. 4l, obtained from three-point bending experiments, further reveal the superior flexibility of MAFAs compared to NP-MAFAs and *p*-NFAs. Additionally, the MAFAs exhibited exceptional flexibility and damage tolerance, maintaining structural integrity without brittle fracture even under extreme 90° bending deformation (Fig. 4m, n and Supplementary Fig. 15).

To elucidate the fundamental mechanisms underlying the exceptional ductility and bendability of MAFAs, FE simulations were subsequently conducted to analyze the composite architecture under flexural loading conditions[1,51,52]. As illustrated in Fig. 4h–k, the MAFAs exhibit enhanced bending stiffness, arising primarily from the additional mechanical constraints imposed by the KTNs, which effectively restrict interlayer shear displacement while promoting efficient stress transfer between adjacent cellular layers[52]. The FE simulations showed a characteristic inverted volcano-shaped stress distribution during maximum bending deformation, demonstrating the efficient stress management of MAFAs system[51]. Specifically, the stress was predominantly accommodated by the high-strength KTNs, along with the nanofibrous cellular layer, which played a critical role in mitigating stress concentration and arresting crack propagation through its energy-absorbing porous architecture. Meanwhile, this hierarchical design enabled effective stress redistribution across multiple length scales, with the KTNs and nanofibrous matrix cooperatively accommodating mechanical loads[52]. Moreover, the MAFAs exhibited enhanced energy storage capacity, with the KTNs accumulating higher specific deformation energy compared to non-reinforced structures under identical displacement conditions. The bending force required to achieve equivalent deformation increased, demonstrating the superior mechanical performance of MAFAs. Critically, these simulations reveal the stress–structure relationships and paving the way for stress-transfer mechanisms governing the exceptional bending resistance and damage tolerance of MAFAs[1]. Overall, this freeze-assisted approach offers a unique synergy that distinctly integrates the advantages of both components, thereby enhancing the mechanical performance of the resulting MAFAs. This integration is essential for simultaneously achieving high strength, stretchability, compressibility, and flexibility.

Thermal stability and thermal insulation applications

Thermal stability represents a fundamental requirement in the design of thermal insulation materials under extreme environmental

https://doi.org/10.1038/s41467-025-62164-4

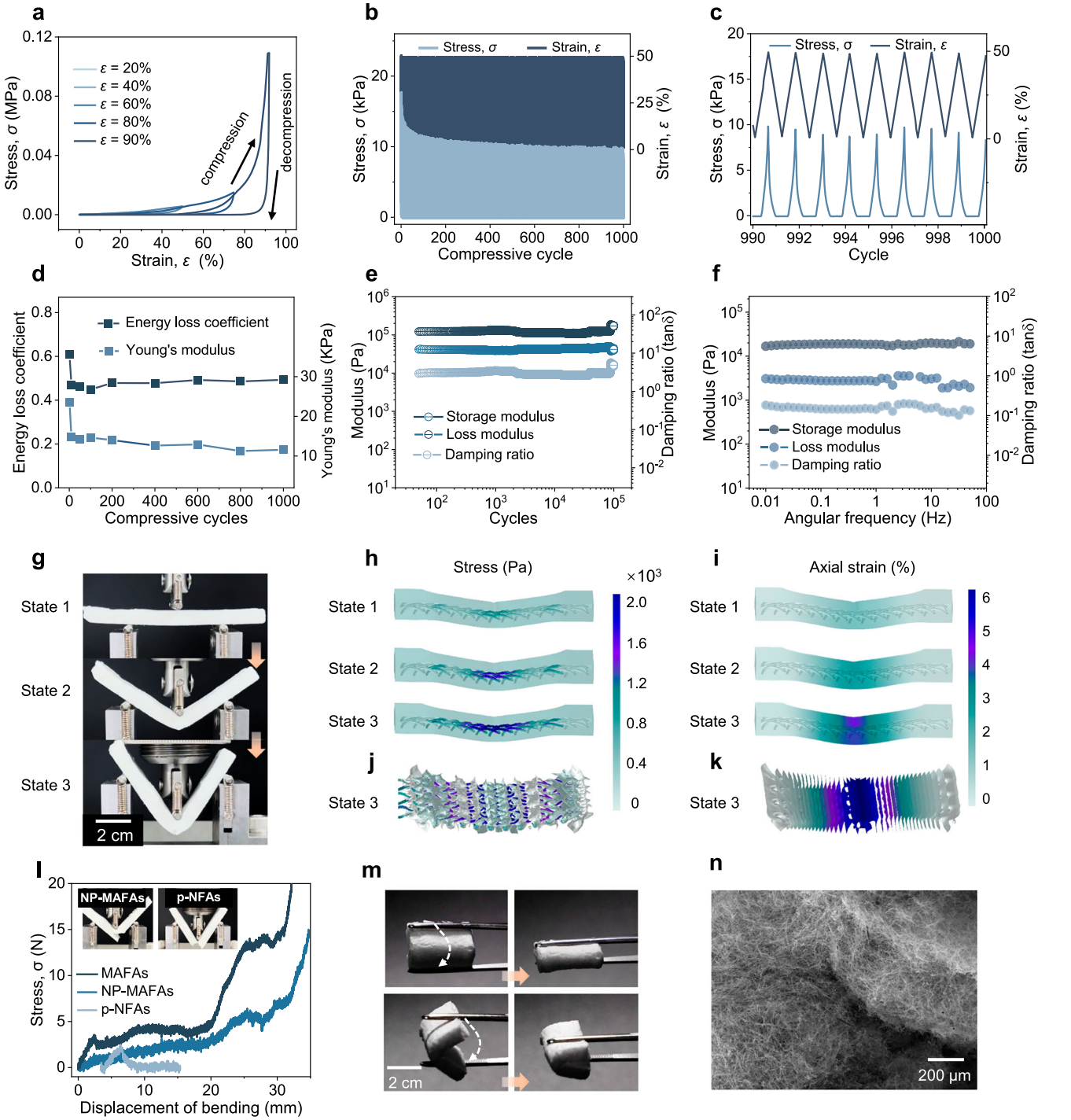


Fig. 4 | Compressive and bending properties of MAFAs. a Compressive σ–ε curves of the MAFAs. **b** Fatigue test curves with an ε of 50% and 1000 cycles. **c** Magnified view of compressive cycles 991-1000. **d** Energy loss coefficient and Young's modulus versus the number of the compression test cycles. **e** The storage modulus, loss modulus and damping ratio of the MAFAs during 10,000 compression-release fatigue cycles with an oscillatory ε of 3%, and a frequency of 5 Hz. **f** The frequency dependence (0.01-100 Hz) of Young's modulus, loss modulus, and damping ratio for MAFAs. **g** Three-point flexure tests on MAFAs using the Instron 34 TM-5 Universal Machine. **h**–**k** Simulation for stress contours of the bending mode of MAFAs. **l** The σ–ε curves obtained from three-point bending experiments performed on MAFAs, NP-MAFAs and *p*-NFAs. (inset of **l**) Three-point flexure tests on NP-MAFAs and *p*-NFAs using the Instron 34 TM-5 Universal Machine. **m, n** Photographs and representative SEM image showing the superior foldability of MAFAs. Source data are provided as a Source Data file.

conditions[53]. For example, in aerospace applications, spacecraft re-entry capsules are subjected to severe aerodynamic heating when entering the atmosphere at high speed (Fig. 5a)[54,55]. To evaluate the thermal insulation performance of MAFAs, we furtherly simulated the temperature distribution in MAFAs under high-temperature conditions. As shown in Fig. 5b, the FE simulations demonstrated that the hierarchically structured ceramic nanofiber cellular network, acting as a gas diffusion barrier, created a steep temperature gradient across the thickness direction of MAFA, revealing its critical role in thermal transport regulation[56]. Moreover, the MAFAs can maintain normal body surface temperature during operation, which is essential in the actual application environment. Mechanistically, the outstanding

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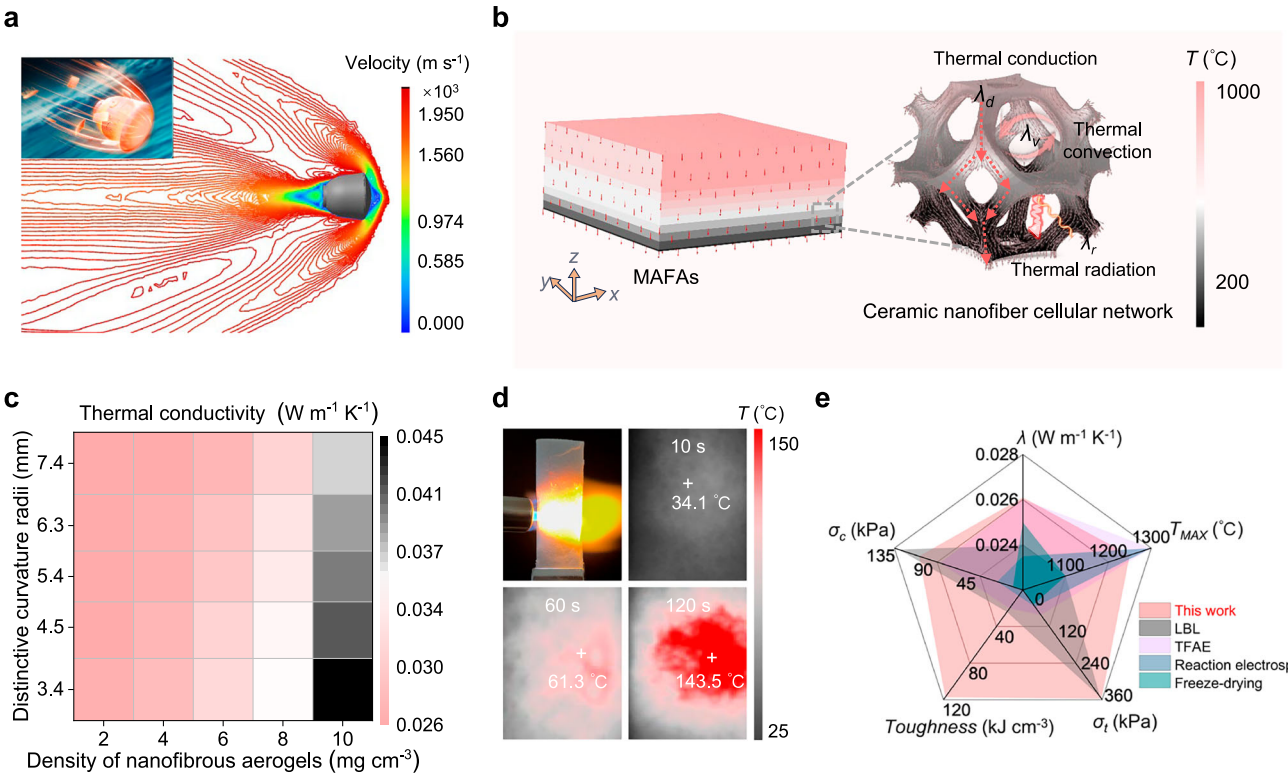


Fig. 5 | Thermal stability and thermal insulation applications of MAFAs. **a** Computational fluid dynamics simulation based on the finite volume method illustrating the velocity distribution nephogram of the reentry capsule within a wind tunnel environment. (insets of **a**) The exterior surface of the bulkhead of reentry capsule is equipped with ablative thermal protection material, which undergoes frictional combustion during atmospheric reentry. **b** Simulated temperature distribution and heat flux directions of integrated MAFAs and a single cell. **c** Heatmap of the thermal conductivity (λ_t) of MAFAs with density of nanofibrous aerogels in the range of 2-10 mg cm^{-3} and topology sizes (Distinctive curvature radii) in the range of 3.4-7.4 mm. Lowest λ_t (pink); highest λ_t (black). **d** Optical and infrared images of the MAFAs after being subjected to a butane blow torch for 120 s. **e** A radar plot comparing the MAFAs and other aerogel-like materials, including SiO_2-ZrO_2 nanofibrous aerogels prepared by the layer-by-layer (LBL) method[58], zircon nanofibrous aerogels prepared by the turbulent flow-assisted electrospinning (TFAE) method[53], mullite aerogels prepared by the direct reaction electrospinning[19], and SiO_2 ceramic aerogels prepared by the freeze-drying method[10]. Source data are provided as a Source Data file.

thermal insulation property of the MAFAs is a result of the low thermal conductivity (λ_t). As shown in the Fig. 5c, the MAFAs possess a minor λ_t of 0.0261 W m^{-1} K^{-1} with a nanofibrous density of 4 g cm^{-3} and a topology size of 7.4 mm at room temperature. Once the nanofibrous density increased to 10 mg cm^{-3} and the topology size increased to 3.4 mm, λ_t increased to 0.045 W m^{-1} K^{-1}, which is attributed to the reduced porosity caused by the increased content of micro-/nanofibers in MAFAs.

Furtherly, a device for extreme temperature tests was constructed to mimic the thermal working conditions. As shown in Fig. 5d, the MAFAs were subjected to a butane flame at approximately 1200 °C, and the real-time temperature variation was captured by an infrared camera. As expected, the MAFAs exhibited a low temperature of 143.5 °C on the backside even after being exposed to constant high temperatures for 120 s, highlighting their extraordinary ability to isolate the thermal environment. Additionally, we similarly compared the thermal insulation properties of MAFAs with those of *p*-NFAs, SiO_2 and Al_2O_3 materials. As shown in the insets of Supplementary Fig. 16, the sample materials were placed on a 350 °C heating plate, and the top surface temperatures (T_s) of the samples were recorded by an infrared camera. Note that for the MAFAs and *p*-NFAs, even after being placed on a 350 °C heating stage for 5 min, the top surface exhibited a relatively low T_s of ~50 °C and then maintained an almost constant T_s for the following 5 min[10,13], which is considerably lower than that of SiO_2 and Al_2O_3 materials. These results indicate the promising thermal insulation performance of our MAFAs. Moreover, the low thermal conductivity, combined with the flame-retardant ability and self-adaptable mechanical properties, enabled the properties of MAFAs to be superior to the current classifications of aerogel-like thermal insulation materials (Fig. 5e and Supplementary Tables 1, 2). Importantly, the assembled MAFAs exhibit exceptional mechanical attributes and thermal stability at high temperatures, thereby rendering them highly sought after for utilization as a thermal insulation layer to reduce heat transfer through the exterior wall of the return capsule (Supplementary Fig. 17). This work highlights the indispensable role of knitted topological frameworks in the efficient design of composite structures with nanoscale precision, contributing to the discovery of more advanced mechanical materials.

Discussion

In summary, a multicomponent and multistructure engineering strategy was proposed by integrating knitting prestressed topological frameworks with ceramic nanofibers aerogels, which were employed to synergistically construct micro-/nanoscale self-adaptable mechanical fibrous aerogels. Notably, the synchronous motion of the prestressed topological units—bearing primary loads and establishing continuous stress-transfer pathways through topological constraints and interlocking—coupled with the sequential buckling of the nanofibrous cellular structures for energy dissipation, endows the MAFAs with remarkable mechanical robustness, including enhanced strength, toughness, and deformation recovery capacity. We expect that this research will provide a new opportunity for the development of mechanical aerogels and other 3D materials for applications as smart and versatile materials.

Methods

Materials

Tetraethyl orthosilicate (TEOS, Si $(OC_2H_5)_4$, 98%, Greagent), Phosphoric acid (H_3PO_4, 85 wt %, Aladdin), Ethanol (EtOH, 99.7%, Greagent), and Poly (vinyl alcohol) 1788 (PVA 1788, Aladdin).

Preparation of flexible ceramic nanofibrous membranes

The SiO_2 nanofibers were synthesized via the sol-gel technique coupled with the electrospinning process. The following three interrelated steps were taken: Initially, the SiO_2 sol was obtained by hydrolyzing of TEOS with H_3PO_4 as catalyst for 10 h, and the mass ratio of TEOS: H_2O: H_3PO_4 = 100: 100: 1.1. The 10 wt% PVA aqueous solution was introduced into the SiO_2 sol in a mass ratio of 1:1 and continued stirring for another 3 h at room temperature to obtain the SiO_2 precursor solution. Subsequently, the SiO_2 precursors were spun using a DEXS-4 spinning machine with an operating voltage of 20-22 kV, a collection length of 15–20 cm, and an ejection speed of 1 mL h^{-1} at a humidity of 35 ± 2% and a temperature of 25 ± 2 °C. Finally, the green SiO_2 nanofibers were immediately vacuum dried at 100 °C for 1 h and calcined in air at 800 °C with a muffle furnace to complete the pyrolysis of the polymer.

Preparation of high-temperature binder

Silica sol was selected as the high-temperature binder. It was synthesized by mixing 2 g of TEOS, 20 g of EtOH, and 0.002 g of H_3PO_4 with 8 mL of deionized water under vigorous stirring at room temperature for at least 30 min.

Preparation of MAFAs

The fabrication of MAFAs involves a sequential process comprising immersion, freezing, freeze-drying, and calcination. Initially, ceramic nanofibers were homogenously dispersed in high-temperature binder (silica sol) to prepare a stable nanofiber/sol suspension (mass ratio of nanofibers: H_2O: silica sol = 1: 200: 2.73). A pre-tensioned KTN framework was then immersed in this suspension for 1 h to ensure uniform infiltration. Following immersion, the prestressed KTN framework was rapidly frozen within the nanofiber/sol dispersion using liquid nitrogen. During cryogenic solidification, phase separation occurred, leading to the directional expulsion and accumulation of nanofiber solids around the growing ice crystals. The frozen composite was subsequently subjected to freeze-drying in a programmable lyophilizer for 48 h, during which the ice crystals sublimed, leaving behind a porous architecture. The final step involved calcination at 800 °C for 1 h in a muffle furnace, which further consolidated the Si-O-Si crosslinking network, ultimately yielding the mechanically adaptive fibrous aerogels (MAFAs).

Characterization

The microstructures of MAFAs were characterized by FE-SEM (S-4800). The tensile and three-point flexural tests mechanical tests were measured by an Instron 34 TM-5 universal machine. The compression mechanical tests were performed using a DMA instrument (TA-Q850). The fast recovery of a compressed MAFAs was recorded by a Fuji Film X-T200 camera. The thermal conductivity of MAFAs at a range of densities was characterized using a hot disk instrument (TPS2500S, Switzerland). An infrared thermal camera (Fluke TiX560) was used to take the infrared images of the MAFAs.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The Source data are provided as a Source Data file. Source data are provided with this paper.

References

1. Zan, G. et al. A biomimetic conductive super-foldable materia. *Matter*. **4**, 3232–3247 (2021).
2. Parale, V. G. et al. Mechanically strengthened aerogels through multi-scale, multi-composition, and multi-dimensional approaches: A review. *Adv. Mater.* **36**, 2307772 (2024).
3. Xu, X. et al. Double-negative-index ceramic aerogels for thermal superinsulation. *Science* **363**, 723–727 (2019).
4. Li, L. et al. Large-scale assembly of isotropic nanofiber aerogels based on columnar-equiaxed crystal transition. *Nat. Commun.* **14**, 5410 (2023).
5. Wang, H. et al. Ultralight, scalable, and high-temperature–resilient ceramic nanofiber sponges. *Sci. Adv.* **3**, e1603170 (2017).
6. Wu, M. et al. Superelastic graphene aerogel-based metamaterials. *Nat. Commun.* **13**, 4561 (2022).
7. Niu, H., Qiang, Z. & Ren, J. Durable, magnetic-responsive melamine sponge composite for high efficiency, in situ oil-water separation. *Nanotechnology* **32**, 275705 (2021).
8. Zhu, P. et al. 3D printed cellulose nanofiber aerogel scaffold with hierarchical porous structures for fast solar-driven atmospheric water harvesting. *Adv. Mater.* **4**, 2306653 (2024).
9. Wang, H., Cheng, L., Yu, J., Si, Y. & Ding, B. Biomimetic Bouligand chiral fibers array enables strong and superelastic ceramic aerogels. *Nat. Commun.* **15**, 336 (2024).
10. Si, Y., Wang, X., Dou, L., Yu, J. & Ding, B. Ultralight and fire-resistant ceramic nanofibrous aerogels with temperature-invariant superelasticity. *Sci. Adv.* **4**, eaas8925 (2018).
11. Yan, Z. et al. Molecular cage-mediated radial gradient porous sponge nanofiber for selective adsorption of a mustard gas simulant. *ACS Appl. Mater. Interfaces* **13**, 47835–47844 (2021).
12. Wang, F. et al. In situ synthesis of biomimeticsilica nanofibrous aerogels with temperature-invariant superelasticity over one million compressions. *Angew. Chem. Int. Ed.* **59**, 8285–8292 (2020).
13. Dou, L. et al. Interweaved cellular structured ceramic nanofibrous aerogels with superior bendability and compressibility. *Adv. Funct. Mater.* **30**, 2005928 (2020).
14. Dou, L. et al. Temperature-invariant superelastic, fatigue resistant, and binary-network structured silica nanofibrous aerogels for thermal superinsulation. *J. Mater. Chem. A.* **8**, 7775–7783 (2020).
15. Zhang, X. et al. Ultrastrong, superelastic, and lamellar multiarch structured ZrO_2-Al_2O_3 nanofibrous aerogels with high-temperature resistance over 1300 °C. *ACS Nano* **14**, 15616–15625 (2020).
16. Wu, F., Liu, Y., Si, Y., Yu, J. & Ding, B. Multiphase ceramic nanofibers with super-elasticity from -96–1600 °C. *Nano Today* **44**, 101455 (2022).
17. Su, L. et al. Anisotropic and hierarchical SiC@SiO_2 nanowire aerogel with exceptional stiffness and stability for thermal superinsulation. *Sci. Adv.* **6**, eaay6689 (2020).
18. Su, L. et al. Ultralight, recoverable, and hightemperature-resistant sic nanowire aerogel. *ACS Nano*. **12**, 3103–3111 (2018).
19. Cheng, X., Liu, Y., Si, Y., Yu, J. & Ding, B. Direct synthesis of highly stretchable ceramic nanofibrous aerogels via 3D reaction electrospinning. *Nat. Commun.* **13**, 2637 (2022).
20. Shen, X. et al. Achieving adjustable elasticity with non-affine to affine transition. *Nat. Mater.* **20**, 1635–1642 (2021).
21. Bertoldi, K., Vitelli, V., Christensen, J. & Van Hecke, M. Flexible mechanical metamaterials. *Nat. Rev. Mater.* **2**, 17066 (2017).
22. Maziz, A. et al. Knitting and weaving artificial muscles. *Sci. Adv.* **3**, e1600327 (2017).
23. Poincloux, S., Adda-Bedia, M. & Lechenault, F. Geometry and elasticity of a knitted fabric. *Phys. Rev. X.* **8**, 021075 (2018).
24. Singal, K. et al. Programming mechanics in knitted materials, stitch by stitch. *Nat. Commun.* **15**, 2622 (2024).

25. Ma, R., Chou, S., Xie, Y. & Pei, Q. Morphological/nanostructural control toward intrinsically stretchable organic electronics. *Chem. Soc. Rev.* **48**, 1741–1786 (2019).
26. Shanahan, W. J. & Postle, R. A. Theoretical analysis of the plain-knitted structure. *Text. Res. J.* **40**, 656–665 (1970).
27. Hepworth, B. & Leaf, G. A. V. 32—The mechanics of an idealized weft-knitted structure. *J. Text. Inst.* **67**, 241–248 (1976).
28. Dusserre, G., Balea, L. & Bernhart, G. Elastic properties prediction of a knitted composite with inlaid yarns subjected to stretching: A coupled semi-analytical model. *Compos. Part A Appl. Sci. Manuf.* **64**, 185–193 (2014).
29. Ka Fai, C. & Tin Yee, L. The shape and dimensions of plain knitted fabric: A fabric mechanical model. *Text. Res. J.* **76**, 777–786 (2016).
30. Kaldor, J. M., James, D. L. & Marschner, S. Simulating knitted cloth at the yarn level. *ACM Trans. Graph.* **27**, 1–9 (2008).
31. Chen, Y. et al. Realistic rendering and animation of knitwear. *IEEE Trans. Vis. Comput. Graph.* **9**, 43–55 (2003).
32. Yan, Z. et al. Ternary-porous conjugated N-halamine nanofibers/graphene aerogels for rechargeable degradation of mustard gas. *Adv. Funct. Mater.* **32**, 2206018 (2022).
33. Wu, K. et al. A prestressing strategy enabled synergistic energy-dissipation in impact-resistant nacre-like structures. *Adv. Sci.* **9**, 2104867 (2022).
34. Polishchuk, I. et al. Coherently aligned nanoparticles within a biogenic single crystal: A biological prestressing strategy. *Science* **358**, 1294–1298 (2017).
35. Hurd, J. C. H. & Doyle, P. J. Fundamental aspects of the design of knitted fabrics. *J. Text. Inst. Proc.* **44**, 561–578 (1953).
36. Peirce, F. T. Geometrical principles applicable to the design of functional fabrics. *Text. Res. J.* **17**, 123–147 (1947).
37. Fu, C. & Wang, X. Prediction of the cohesive strength for simulating composite delamination by a micro-mechanical model based on random RVE. *Compos. Struct.* **262**, 113343 (2021).
38. Wang, F., Si, Y., Yu, J. & Ding, B. Tailoring nanonets-engineered superflexible nanofibrous aerogels with hierarchical cage-like architecture enables renewable antimicrobial air filtration. *Adv. Funct. Mater.* **31**, 2107223 (2021).
39. Zhang, X. et al. Mechanically interlocked aerogels with densely rotaxanated backbones. *J. Am. Chem. Soc.* **144**, 11434–11443 (2022).
40. Ducrot, E., Chen, Y., Bulters, M., Sijbesma, R. P. & Creton, C. Toughening elastomers with sacrificial bonds and watching them break. *Science* **344**, 186–189 (2014).
41. Takahashi, R. et al. Creating stiff, tough, and functional hydrogel composites with low-melting-point alloys. *Adv. Mater.* **30**, 1706885 (2018).
42. King, D. R. Macroscale double networks: Highly dissipative soft composites. *Polym. J.* **54**, 943–955 (2022).
43. Shaikeea, A. J. D., Cui, H., O'Masta, M., Zheng, X. & Deshpande, V. S. The toughness of mechanical metamaterials. *Nat. Mater.* **21**, 297–304 (2022).
44. Zheng, Z. et al. Bio-inspired integrated auxetic elastomers constructed by dual dynamic interfacial healing strategy. *Adv. Mater.* **35**, 2304631 (2023).
45. Cheng, Y. et al. Super-elasticity at 4 K of covalently crosslinked polyimide aerogels with negative Poisson's ratio. *Nat. Commun.* **12**, 4092 (2021).
46. Wang, Y. et al. Highly compressible and environmentally adaptive conductors with high-tortuosity interconnected cellular architecture. *Nat. Synth.* **1**, 975–986 (2022).
47. Guo, X., Dong, X., Zou, G., Gao, H. & Zhai, W. Strong and tough fibrous hydrogels reinforced by multiscale hierarchical structures with multimechanisms. *Sci. Adv.* **9**, eadf7075 (2023).
48. Dou, L. et al. Hierarchical cellular structured ceramic nanofibrous aerogels with temperature-invariant superelasticity for thermal insulation. *ACS Appl. Mater. Interfaces* **11**, 29056–29064 (2019).
49. Zu, G., Kanamori, K., Maeno, A., Kaji, H. & Nakanishi, K. Superflexible multifunctional polyvinylpolydimethylsiloxane-based aerogels as efficient absorbents, thermal superinsulators, and strain sensors. *Angew. Chem. Int. Ed.* **130**, 9870–9875 (2018).
50. Chen, Y. et al. Multi-layered cement-hydrogel composite with high toughness, low thermal conductivity, and self-healing capability. *Nat. Commun.* **14**, 3438 (2023).
51. Chen, K. et al. Graphene oxide bulk material reinforced by heterophase platelets with multiscale interface crosslinking. *Nat. Mater.* **21**, 1121–1129 (2022).
52. Wang, B. et al. Folding large graphene-on-polymer films yields laminated composites with enhanced mechanical performance. *Adv. Mater.* **30**, e1707449 (2018).
53. Guo, J. et al. Hypocrystalline ceramic aerogels for thermal insulation at extreme conditions. *Nature* **606**, 909–916 (2022).
54. Gu, D. et al. Material-structure-performance integrated laser-metal additive manufacturing. *Science* **372**, eabg1487 (2021).
55. Fankong, M. et al. Research on thermal control system optimization of China new generation manned spacecraft reentry capsule. *Spacecr. Recov. Remote Sens.* **42**, 10–21 (2021).
56. Liu, Y. et al. Scalable-produced 3D elastic thermoelectric network for body heat harvesting. *Nat. Commun.* **14**, 3058 (2023).
57. Zhang, X., Cheng, X., Si, Y., Yu, J. & Ding, B. Elastic and highly fatigue resistant ZrO_2-SiO_2 nanofibrous gel with low energy dissipation for thermal insulation. *Chem. Eng. J.* **433**, 133628 (2022).
58. Zhang, X., Cheng, X., Si, Y., Yu, J. & Ding, B. All-ceramic and elastic aerogels with nanofibrous-granular binary synergistic structure for thermal superinsulation. *ACS Nano*. **16**, 5487–5495 (2022).
59. Su, L. et al. Strong yet flexible ceramic aerogel. *Nat. Commun.* **14**, 7057 (2023).

Acknowledgements

This work is supported by the Ministry of Science and Technology of China (No. 2023YFC3011705), the National Natural Science Foundation of China (No. 52373281 and 92371110), Fundamental Research Funds for the Central Universities and Graduate Student Innovation Fund of Donghua University (No. CUSF-DH-D-2022025), Weiqiao Science Foundation (No. H2872302 and H2872303).

Author contributions

H.W. and Y.S. conceived and designed the study. H.W. organized the data and wrote the manuscript. L.C. and J.Y. supervised the project. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-025-62164-4.

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Peer review information *Nature Communications* thanks Shanyu Zhao and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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